B.SC. MAJOR PROJECT — PLAIN-LANGUAGE SUMMARY

MedXVLM

An AI System for Reading Chest X-Rays — What Was Built, How Well It Actually Works, and How It Compares

Image + Text AI | Chest X-Rays | Compared Against 2 Published AI Models | Rebuilt & Re-tested Honestly

Prepared 2026-09-26 (updated with confidence intervals and the image-encoder comparison) · Kartik Singh Rana

1. What This Project Is, In One Paragraph

MedXVLM is a computer program that looks at a chest X-ray image and does three things a radiologist does: it learns to connect what it *sees* in the image with what a doctor *writes* about it, it flags which specific problems (like an enlarged heart or fluid around the lungs) are likely present, and it drafts a written report describing the findings. The project's goal was not just to build this pipeline, but to test it rigorously — including testing it against two real, published research AI systems (called **RadZero** and **CheXOne**) on the same images, so the results are a fair, honest picture of what actually works and what doesn't.

> **Why this matters:** Chest X-rays are one of the most common medical scans in the world, and there aren't enough radiologists everywhere to read them all quickly. AI systems like this one are being researched as a way to give a "second opinion" fast, or highlight images that need urgent attention.

The three-stage pipeline that was built

STAGE 1

Learning to Connect Pictures and Words

The model is shown thousands of X-ray images paired with their real reports, and learns which visual patterns go with which words (e.g. what "cardiomegaly" looks like on film).

STAGE 2

Spotting Specific Problems

Using what it learned in Stage 1, the model is trained to tick "yes/no" boxes for 14 standard conditions (e.g. Pneumonia, Fracture, Fluid around the lung) for a new, unseen image.

STAGE 3

Writing the Report

The model is connected to a small language model (like a mini version of ChatGPT) so it can write a findings paragraph in full sentences, not just a checklist.

All of this was trained on a real, public dataset of **3,826 usable chest X-ray studies and 7,426 images** from Indiana University (a well-known open medical imaging dataset), split fairly into training data, a validation set (used only for tuning), and a completely separate test set the model never saw during training — that test set is what every "how good is it" number below is measured on.

2. Does It Spot Diseases? — The Honest Answer

The main way to measure "how good is this at spotting disease" is a score called **AUROC** (think of it as a 0–100% score for how well the model tells sick from healthy). A quick way to read it:

> **50% = pure guessing** (a coin flip). **100% = perfect.** Real hospital-grade AI systems for this kind of task usually score somewhere in the **80–90%** range.

First result: the full pipeline clearly beat a simple baseline



> **But here's the honest catch, found during later testing:** that 87% was measured in a way that wasn't fair. The model was also given a copy of the *real doctor's report text* for the same X-ray as an extra hint while it made its prediction — like giving a student the answer key during the test. Re-tested **without** the hint (image only, exactly how a doctor would use it), the score dropped to **51% — essentially random guessing**.

What we found and fixed

- **Thousands of training labels were wrong.** The reports are turned into yes/no labels by a rule-based reader. In a sentence like "*No consolidation, pleural effusion, **or pneumothorax***" the word "no" didn't reach the last item, so clean X-rays were labelled as *having* a collapsed lung. After the fix, the pneumothorax rate fell from 10% to 0.8% (a clinically plausible level) and fluid-around-the-lung from 20% to 5%. An independent check: an outside model (RadZero) agreed far more with the corrected labels than the old ones.
- **The classifier was retrained image-only**, so it can no longer read the answer out of the report.
- **Front-view images only**, and an image encoder pre-trained specifically on chest X-rays (a public, openly-licensed one).





All bars: the same 377 never-seen-before front-view test X-rays, the same corrected labels, no hint to anyone. Averaged over the 8 conditions shared with RadZero. "One training run" is the average of four runs with different random seeds (84% ± 1.5); an earlier version of this document quoted 86% for a single run, which turned out to be the luckiest of the four.

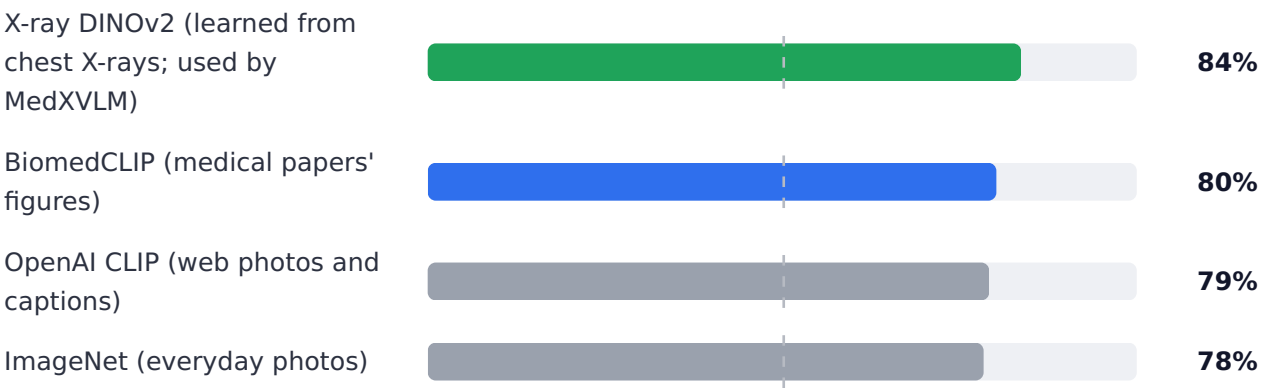
> **The numbers to quote: 84% ± 1.5 for a single model, 87% (0.868 AUROC) when four independently-trained copies are averaged — the same score as RadZero, a published system trained on roughly 100× more data, on identical images.** Averaging several models is a standard, honest technique; it also makes the model slightly better on X-rays from a different hospital (72% vs 70%). Most of the recovery came from cleaning the labels, front views and the X-ray-specific encoder; the project's own attention-based design adds a further 5–8 points over a plain classifier on the same data (78–80%).

Is "equal to RadZero" a real tie, or luck?

A statistical check (re-drawing the 377 test X-rays thousands of times and re-scoring both systems on each draw) puts the difference between the 4-model average and RadZero at **0.000, with a 95% range of −0.033 to +0.033**. In plain terms: on these images the two are indistinguishable, and any real gap larger than about 3 points in either direction is ruled out. The same check confirms that the project's own design, the per-finding attention step, adds a real 5–8 points over a plain classifier.

Does it matter what the image encoder learned beforehand?

One of the project's four research questions asked whether an image encoder pretrained on *medical* images beats one pretrained on everyday photos. The same classifier was retrained on four encoders:



Disease-spotting score (AUROC) on the same 377 test X-rays. X-ray DINOv2 is the average of 4 training runs, the others one run each.

The X-ray-trained encoder wins, and the gap grows where it matters: on the unseen hospital (70% vs 63% for ImageNet) and above all for the heatmaps, where the ImageNet encoder's maps land on the radiologist's box only **13%** of the time against **32%**. The specific comparison the proposal named, medical BiomedCLIP vs general CLIP, goes the predicted way but is too close to call with one training run each.

3. Does It Write Reports? — Before and After the Retrain

Stage 3 writes a "findings" paragraph for an X-ray. It is scored by extracting the medical claims from the AI's text (e.g. "no fracture", "enlarged heart") and checking them against the true findings — a **clinical-efficacy score** from 0 to 1 — rather than judging whether the text merely *sounds* like a report.

The original model had collapsed into one sentence

Tested on 150 different X-rays, the first version produced **only 5 different reports in total, and 106 of the 150 were the identical text** — a generic "everything looks normal" paragraph, even for images whose real reports described an enlarged heart or a fracture. It had learned that "normal" is the safe answer (70% of training reports are normal) and stopped looking at the image.

What the retrain changed

- Built on the rebuilt, X-ray-specific image encoder from Stage 2 — the one that can actually tell findings apart.
- Front views only, cleaned report text, more image detail passed to the language model.
- **Abnormal cases shown to the model as often as normal ones during training**, so "say normal" stops being the cheapest strategy.

System (same 70 front-view test X-rays)	Clinical-efficacy score	Different reports produced (of 150)	Invents catheters / lines that aren't there
MedXVLM, retrained — beam-search decoding (the demo's default)	**0.176**	14 (of 70 front views)	3% of cases
MedXVLM "v2c" — told the classifier's findings before writing	0.165	13 (of 70)	**0%**
Ceiling: the classifier's findings written out as a bare list	*0.234*	*13 (of 70)*	*0%*
	0.152	26 (of 70)	10% of cases

MedXVLM, retrained — plain (greedy) decoding			
CheXOne (outside AI, off-the-shelf)	0.149	36 (of 70)	0%
MedXVLM, original	0.078	5 (of 70)	1%

One more experiment (v2c): since the attention study (section 5) showed the writer barely looks at the image, the classifier's findings were written into its instructions — "Classifier findings: Cardiomegaly, Pleural Effusion." That **eliminated invented equipment entirely (0%)** and made the report an honest narration of what the classifier found. It did *not* raise the score (0.165 vs 0.176 — a tie within the measurement noise of ±0.10 on 70 images), and the reason is instructive: even a perfect narrator of the classifier can only score 0.234 on these images, because the classifier itself flags a finding on just 18 of the 70. The report writer is now limited by the classifier's ability to spot rare findings, not by its writing. Take the findings out of the instructions and the same model collapses back to one sentence for every image — proof that the image reaches the report only through the classifier.

The good: the collapse is fixed, the score more than doubled, and on this measure the project's model now **edges ahead of CheXOne** — a 3-billion-parameter system from a Stanford lab. It correctly writes "Cardiomegaly" for enlarged hearts the old model always called normal. A follow-up experiment found that **how the text is decoded matters as much as how the model was trained**: switching from picking the single most likely next word ("greedy") to considering several candidate sentences at once ("beam search") cut the invented-equipment problem from 10% to 3% and raised the score to 0.176 — with no retraining. A separate test that reduced the "show abnormal cases more often" setting made the model collapse completely again (one identical sentence for all 150 images), which proves that setting was essential.

> **The honest remaining problem:** in about 1 in 30 front-view cases (1 in 10 with plain decoding) the retrained model **invents medical equipment or surgery that isn't in the image** — "IJ catheter tip at the SVC", "surgical clips from prior lobectomy". This happened because abnormal training reports mention lines and tubes four times more often than normal ones, so when we forced the model to pay attention to abnormal cases it learned "abnormal ⇒ mention a catheter" as a shortcut. CheXOne never does this. So the retrain traded one failure (always saying normal) for a smaller but more dangerous one (making things up), which better decoding then mostly contained. Neither model is anywhere near usable for real report drafting on ~3,000 training studies; a score of 0.18 means most

> specific findings are still missed, and the model still gives the same "normal" paragraph for about 6 in 10 front-view images.

Two traditional text-similarity scores (BLEU, ROUGE) were also computed and slightly favour the *original, collapsed* model — because its boilerplate happens to match the most common reference wording. That is a known weakness of those metrics for radiology reports, which is why the clinical-efficacy score is used to rank the systems here.

4. How Does It Compare to Real Published AI Systems?

RadZero (2025)

A small, specialised AI (about 200 million parameters) that, given an X-ray and a disease name, says how likely that disease is present and can highlight *where* it's looking. Trained on ~350,000 X-ray/report pairs. Its code and weights are public.

CheXOne (2026)

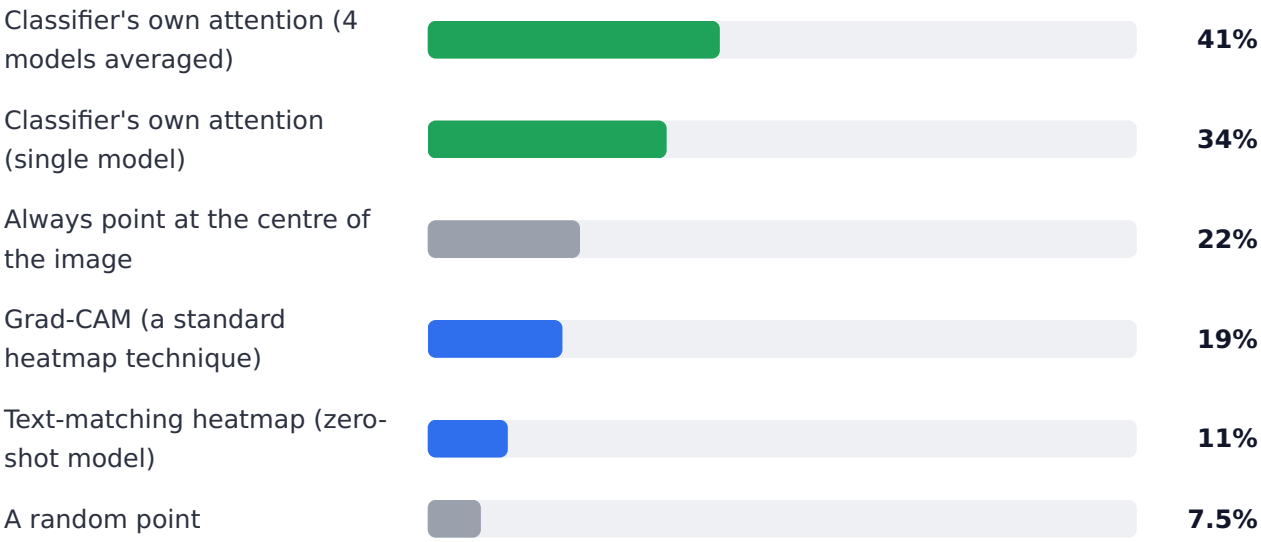
A much larger AI (3 billion parameters, built on chatbot-style technology) from a Stanford-affiliated group. Its specialty is writing full radiology reports with a visible reasoning trace. Trained on a purpose-built 14.7-million-example dataset. Weights are public; its own benchmarks need restricted hospital data.

Task	Outside system	Its score	MedXVLM (rebuilt)	Same images & labels?
Spotting diseases (AUROC, 8 shared conditions, 377 test X-rays)	RadZero	87%	87% (4-model average); 84% ± 1.5 single	Yes — fair comparison
Writing reports (clinical-efficacy, 70 front-view test X-rays)	CheXOne	0.149	0.176	Yes — fair comparison
Describing a disease in words and matching it to an image ("zero-shot", no labels used)	RadZero	87%	73%	Yes, but RadZero trained on ~100× more text

Both outside systems were **actually run** on this project's test images — these are real measured numbers, not copied from their papers. The reverse (running MedXVLM on *their* benchmark datasets) was not possible: those datasets require credentialed hospital-data access. Matching RadZero (with a 4-model average) and edging past CheXOne, on ~100× less training data, is a strong result for a project of this size — but note both report scores are low in absolute terms, and the "zero-shot" row shows where the data gap really bites: describing a finding in words and matching it to an image reached 73% only after training specifically on the 1,683 sentences that mention a finding; RadZero has hundreds of thousands.

5. Can It Show *Where* It Looked? — Explainability, Measured

A model that says "pleural effusion, 80%" is more useful — and more trustworthy — if it can also point at the part of the X-ray that made it say so. The project produces such heatmaps in several ways, and until now they had only been eyeballed. They have now been **scored against 984 boxes that radiologists drew around real findings** on 880 X-rays from a different hospital (the public NIH ChestX-ray14 collection). The test is simple: does the heatmap's hottest point fall inside the radiologist's box?



Share of 984 radiologist boxes whose finding the heatmap's hottest pixel lands inside. "Centre" scores 22% only because enlarged-heart boxes cover the middle of every image (97% for that finding alone); leaving that finding out, centre drops to 8% and the classifier's attention to 29%.

The classifier's heatmaps point at real findings. For fluid around the lung, collapsed lung tissue, lung opacities and pneumonia the hottest point is inside the radiologist's box 33–62% of the time — five to ten times more often than chance. Averaging four models helps here too (34% → 41%). This is the first validated explainability result in the project.

Where the model is weak, the heatmaps are meaningless. Pneumothorax (collapsed lung) heatmaps are no better than chance — the same condition the classifier itself scores worst on, because it had almost no examples. Two textbook techniques the proposal called for were built and turned out *worse* than the model's own attention: Grad-CAM (19%) and a generic "what is salient" map (6%,

> i.e. chance). One planned research question — whether a finer-grained training signal in Stage 1 makes heatmaps more accurate — got a clear answer: **no measurable effect at this data size** (5% with it, 4% without, both below random).

And the report writer? A separate check traced, for every sentence the report writer produces, how much of its "attention" went to the image versus to the words it had already written. The answer: about **1%**. Even sentences that assert a finding give the image only ~3% of their attention, and every sentence in a report re-uses the same glance at the image (the maps for different sentences agree at 0.91 on a 0–1 scale). Where the writer does name a finding that a radiologist boxed, that glance is on the box 55% of the time (centre baseline 32%) — so the image information is there; the writer just barely uses it. That is a measured reason for why the report writer stays weak — and it points to the fix: give it the classifier's findings as text, which is the channel it actually uses.

6. Key Takeaways

- **The full 3-stage pipeline is built, trained on real data, and runs end-to-end** — nothing here is simulated or estimated.
- **The first headline number (87%) was misleading** — measured with the answer key; the honest image-only score was 51%. Catching this by testing the hard way is one of the project's most important outcomes.
- **Root causes were found and fixed**: a label-reading bug that mislabelled thousands of normal X-rays, a training setup that leaked the answer, side-view images diluting the data, a non-specialised image encoder — and, for report writing, a "safe answer" collapse.
- **After the rebuild and the overnight follow-ups:** disease-spotting at **84% ± 1.5 per model, 87% averaged over four — equal to RadZero**; report-writing clinical-efficacy **0.176, ahead of CheXOne's 0.149**; both on identical test images with identical corrected labels. An earlier single-run figure of 86% was the best of four seeds and should not be quoted alone.
- **All four research questions now have measured answers.** The newest: the image encoder's pretraining matters, a little for accuracy and a lot for whether the heatmaps point at the right place.
- **The report writer's ceiling is the classifier:** telling it the classifier's findings removes invented equipment completely (0%) but cannot lift the score above what the classifier already knows (a hard cap of 0.234 on the test images); better reports now need better rare-finding detection, i.e. more data.
- **Explainability is now measured, not assumed:** the classifier's heatmaps land inside the radiologist's box on **41%** of 984 real boxes (random: 7.5%); the report writer, by contrast, barely looks at the image at all.
- **Open problems, stated plainly:** the report writer still gives the same "normal" paragraph for ~6 in 10 front-view images and invents equipment in ~3%; it misses most specific abnormal findings; zero-shot matching sits at 73% vs 87% for RadZero (a data-scale gap); and after the label fix some conditions have very few true test cases (one pneumothorax, five consolidations), so their individual scores are not meaningful.

What would come next

1. Done: the report writer was given the classifier's findings and checkpoints were chosen by generation quality — it fixed hallucination, and showed the remaining limit is the classifier's recall on rare conditions. The next real step is more labelled examples of those conditions.

2. The external-hospital test is done (NIH ChestX-ray14: 71% single model, 72% averaged); a further step would be a second external dataset with different labelling.
3. Get more true examples of the rare conditions so their individual scores become meaningful.

7. Plain-Language Glossary

AUROC

A 0–100% score for how well a model tells "disease present" apart from "disease absent." 50% = coin-flip guessing; 100% = perfect.

Clinical-efficacy score

For written reports: extracts the specific medical claims from the AI's text and checks them against the true findings, rather than judging fluency. 0 to 1; higher is better.

Heatmap / pointing game

A heatmap colours the parts of the image the model relied on. The pointing game scores it: 1 if the hottest pixel is inside the box a radiologist drew around the finding, 0 otherwise.

Hallucination

When a language model states something specific (here: a catheter, a surgical clip) that has no basis in its input. Dangerous in medical text because it reads as confident fact.

Blind / image-only test

Testing the model using *only* the X-ray image — no extra hints, exactly like a genuinely new patient.

Held-out test set

Data set aside and never shown to the model during training or tuning, used only at the end to get an honest score.

Parameters

Roughly, the "size" of an AI model — how many internal adjustable numbers it has. More generally means more capacity, but also more data and compute needed to train it well.

Full technical detail and every source file behind every number: `docs/status_report.md` (sourced inventory), `docs/v2_rebuild.md`, `docs/cross_dataset_evaluation.md`, `docs/external_baseline_reproduction.md`, `docs/training_summary.md`.